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Exercise 1: Implementing the Singleton Pattern

```
class Logger {
    private static Logger log;

    private Logger() {
        System.out.println("Logger");
    }

    public static Logger getLogger() {
        if (log == null) {
            log = new Logger();
        }
        return log;
    }

    public void log(String message) {
        System.out.println(message);
    }
}

public class Main {
    public static void main(String[] args) {

        Logger log1 = Logger.getLogger();
        Logger log2 = Logger.getLogger();

        log1.log("First Log message");
        log2.log("Second Log message");

        if (log1 == log2) {
            System.out.println("both instances are the same");
        } else {
            System.out.println("both instances are not the same");
        }
    }
}
```

```
Logger
First Log message
Second Log message
both instances are the same

=== Code Execution Successful ===
```

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Exercise 2: Implementing the Factory Method Pattern

Document.java

```
public interface Document {  
    void open();  
}
```

DocumentFactory.java

```
public abstract class DocumentFactory {  
    public abstract Document createDocument();  
}
```

WordDocument.java

```
public class WordDocument implements Document {  
    @Override  
    public void open() {  
        System.out.println("Opening Word document");  
    }  
}
```

PdfDocument.java

```
public class PdfDocument implements Document {  
    @Override  
    public void open() {  
        System.out.println("Opening PDF document");  
    }  
}
```

ExcelDocument.java

```
public class ExcelDocument implements Document {  
    @Override  
    public void open() {  
        System.out.println("Opening Excel document");  
    }  
}
```

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WordFactory.java

```
public class WordFactory extends DocumentFactory {
    @Override
    public Document createDocument() {
        return new WordDocument();
    }
}
```

PdfFactory.java

```
public class PdfFactory extends DocumentFactory {
    @Override
    public Document createDocument() {
        return new PdfDocument();
    }
}
```

ExcelFactory.java

```
public class ExcelFactory extends DocumentFactory {
    @Override
    public Document createDocument() {
        return new ExcelDocument();
    }
}
```

Main.java

```
public class Main {
    public static void main(String[] args) {
        DocumentFactory wordFactory = new WordFactory();
        Document word = wordFactory.createDocument();
        word.open();

        DocumentFactory pdfFactory = new PdfFactory();
        Document pdf = pdfFactory.createDocument();
        pdf.open();

        DocumentFactory excelFactory = new ExcelFactory();
        Document excel = excelFactory.createDocument();
        excel.open();
    }
}
```

Opening Word document

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Opening PDF document

Opening Excel document